

PERFORMANCE OF FIXED WIDTH CONFIDENCE INTERVALS UNDER TYPE II ERRORS: THE EXPONENTIAL CASE

H.I. Hamdy

College of Administrative Sciences, Department of Quantitative Methods and
Information, Kuwait University, P.O. Box 5486, 13055 Safat, KUWAIT

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Summary: This article sheds light on the performance of a triple sampling fixed width confidence interval as it pertains to estimating the location parameter of the exponential distribution. The ability of such a confidence interval to identify shifts in the exponential location parameter is thoroughly investigated. Fixed sample size confidence intervals are also evaluated. The study showed that the fixed width confidence interval is unable to detect shifts from the true parameter, in general, and it is very insensitive to right-shifts in particular. Results concerning comparisons with the fixed sample size confidence interval are also presented.

1. Introduction

In this study we consider the two-parameter exponential distribution of the form

$$f(t; \mu_0, \sigma) = \sigma^{-1} e^{-\sigma^{-1}(t-\mu_0)} I(t > \mu_0) - \infty < \mu_0 < \infty, \sigma > 0, \quad (1.1)$$

where $I(\cdot)$ is the usual indicator set function. The location parameter μ_0 and the scale parameter σ are assumed unknown.

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Having observed a random sample $X_1, X_2, \dots, X_n$ of size $n (\geq 2)$ from the distribution in (1.1) we use the sample measures $X_{n(1)} = \min_{1 \leq i \leq n} X_i$ and $\hat{\sigma}_n = (n-1)^{-1} \sum_{i=1}^n (X_i - X_{n(1)})$ to estimate μ_0 and σ , respectively. Suppose further that it is required to estimate μ_0 with a confidence interval such that the precision $d (> 0)$, and the coverage probability, $100(1 - \alpha)\%$, of the interval are predetermined. Naturally, we would propose the interval $I_n = (X_{n(1)} - d, X_{n(1)})$ for μ_0 . It is well known that the optimal (fixed) sample size required to satisfy the above demands should be chosen such that $n \geq n^*$, where

$$n^* = \frac{a\sigma}{d} \text{ and } a = \ln(\alpha^{-1}). \quad (1.2)$$

If σ were known, a fixed sample size procedure based on n^* in (1.2) to construct I_{n^*} would be employed. Since σ is unknown, multistage sampling procedures (two stage, purely sequential, three stage, and accelerated sequential procedures) can be used to estimate n^* via estimation of σ and hence construct the required interval for μ_0 . By so doing, the sample size becomes random, while the precision of estimation d and the coverage probability are in control.

The problem of constructing a fixed width confidence interval for the unknown location parameter μ_0 of the two-parameter exponential distribution has been discussed by many authors including Basu (1971), Mukhopadhyay (1974), Swanepoel and van Wyk (1982), Hamdy (1983), Mauromoustakos (1984), Hamdy *et al.* (1988), Mukhopadhyay (1988), Hamdy *et al.* (1989), Son and Hamdy (1990) and Hamdy *et al.* (1996).

The literature in this area focuses primarily on two issues, the closeness of the triple sampling coverage probability to the nominal value, and the characteristics of the sample size. Further, a natural question is whether the interval reacts to shifts in the location parameter μ_0 . In other words, we should inquire about the ability of the fixed precision confidence interval to detect any departure in the true parameter μ_0 . In the case of the normal distribution, the analogous problem has been studied by Costanza *et al.* (1995). The main goal of the present work is to investigate this issue in depth while estimating the exponential location parameter.

2. Three stage-sampling procedure

Hall (1981) introduced the triple sampling procedure to monitor the process of estimating n^* against the possibility of over-sampling at any stage. The details of the procedure are quite simple and can be described as follows. According to certain decision criteria, estimation is accomplished in three consecutive sampling stages, namely the pilot study, the main study and the fine tuning. The sampling procedure starts by taking an initial random sample of size $m (\geq 2)$ from the population in (1.1) and computing $X_{m(1)}$ and $\hat{\sigma}_m$ to initiate the estimation process. Then, we choose and fix a constant $\gamma \in (0, 1)$ which represents the fraction of n^* to be estimated during the main study. The decision whether to stop sampling or to continue the second stage is taken according to the following decision rule:

$$N_1 = \max\{m, [\gamma a \hat{\sigma}_m / d] + 1\}, \quad (2.1)$$

where $[x]$ is the largest integer less than or equal to x . If the decision is to continue sampling, an additional sample of size $N_1 - m$ is taken and augmented with the previous sample to compute $X_{N_1(1)}$ and $\hat{\sigma}_{N_1}$. The final stage sample is determined by the following stopping rule:

$$N = \max\{N_1, [a\hat{\sigma}_{N_1}/d] + 1\}. \quad (2.2)$$

If the decision is to proceed, we observe $N - N_1$ more samples and terminate the sampling process. Once N is determined, we propose $X_{N(1)}$ and $\hat{\sigma}_N$ as estimates of μ_0 and σ , respectively. Hence, we construct the interval $I_N = (X_{N(1)} - d, X_{N(1)})$ for μ_0 . The asymptotic characteristics of the triple sampling are summarized in the following theorem.

THEOREM 1. In the context of the three-stage procedure (2.1)–(2.2), we have as $d \rightarrow 0$,

- (i) $E(N) = n^* + 1/2 - \gamma^{-1} + o(1)$;
- (ii) $Var(N) = \gamma^{-1}n^* + o(d^{-1})$;
- (iii) $E|N - n^*|^3 = o(d^{-2})$;
- (iv) $P(\mu_0 \in I_N) = 1 - \alpha - a\alpha(2\gamma n^*)^{-1}(a - \gamma + 2) + o(d)$.

The proof of Theorem 1 can be found in Hamdy *et al.* (1989), therefore we omit details.

REMARK 1. From a result of Anscombe (1952) (see also Bhattacharya and Mallik (1973) and Hall (1981, p. 1231)) it follows that $Q = (\gamma/n^*)^{1/2}(N - n^*) \xrightarrow{\ell} N(0, 1)$ as $d \rightarrow 0$, hence $Q^2 \xrightarrow{\ell} \chi^2_{(1)}$ as $d \rightarrow 0$. In sequel, the uniform integrability of Q^r for $r \geq 1$ (see Woodroffe (1977), Theorem (2.3)) yields $E(N - n^*)^2 = \gamma^{-1}n^* + o(d^{-1}) = Var(N)$ as $d \rightarrow 0$.

The following Theorem 2 extends the result of Theorem 1 to obtain second-order approximation of moments of a bounded real-valued function $g(> 0)$ of N .

THEOREM 2. Let $g(> 0)$ be a continuously differentiable, real-valued function in a neighbourhood of n^* and N defined by the rule (2.2), if $\sup_{n \geq m} |g'''(n)| = O(|g'''(n^*)|)$, then as $d \rightarrow 0$ we have

$$Eg(N) = g(n^*) + (2\gamma)^{-1}\{(\gamma - 2)g'(n^*) + n^*g''(n^*)\} + o(d^{-2} |g'''(n^*)|)$$

where E denotes expectation and prime means derivative.

PROOF OF THEOREM 2

Taylor expansion of $g(N)$ around n^* yields

$$g(N) = g(n^*) + (N - n^*)g'(n^*) + (1/2)(N - n^*)^2g''(n^*) + r(d). \quad (2.3)$$

where $r(d)$ is the remainder term. Then, take the expectation of (2.3) and make use of (i) and (ii) of Theorem 1 and Remark 1 to obtain

$$Eg(N) = g(n^*) + (2\gamma)^{-1}(\gamma - 2)g'(n^*) + (2\gamma)^{-1}n^*g''(n^*) + Er(d).$$

We turn to evaluate $Er(d)$ as follows:

$Er(d) \leq (1/6)E |(N - n^*)^3g'''(\nu)|$, where ν is a random variable lying between N and n^* . Therefore, $Er(d) \leq (1/6) \sup_{n \geq m} |g'''(n)| E |N - n^*|^3 = o(d^{-2} |g'''(n^*)|)$ by (iii) of Theorem 1 and the assumption that g''' is bounded. Theorem 2 will be used to study the ability of I_N to detect shifts from μ_0 in the following section.

3. Detecting departure from μ_0

Since the confidence interval contains all those parameter values that would not be rejected if they were used as null hypothesis values, careful consideration of the ability of the interval to detect shifts in the true parameter should be made to ensure that a credible inference will be obtained.

Consider the null hypothesis $H_0 : \mu = \mu_0$, which asserts that no shift in the true parameter μ_0 has occurred, against the alternative hypothesis $H_1 : \mu = \mu_1 = \mu_0 - kd$, which asserts that the parameter value differs from μ_0 by a "distance" k measured in units of d . The probability of not detecting a k shift in the true parameter μ_0 can be measured by the corresponding Type II error β (β -risk), which is the conditional probability of not detecting such shift when, in fact, such a shift actually took place. Let β_t denotes the β -risk associated with the triple sampling fixed width confidence interval. Hence,

$$\begin{aligned}\beta_t &= P(\mu_0 \in I_N \mid H_1) \\ &= \sum_{n=m}^{\infty} P(dk \leq X_{N(1)} - \mu_1 \leq d(1+k), N=n)\end{aligned}$$

The event $N = n$ depends only on $\hat{\sigma}_N$ from the rule (2.2) for $n = m, m+1, \dots$. On the other hand $\hat{\sigma}_n$ and $X_{n(1)}$ are independent by Basu (1955) Theorem. It follows that

$$\beta_t = Ee^{-Ndk\sigma^{-1}} - Ee^{-Nd(k+1)\sigma^{-1}}.$$

Finally, we apply Theorem 2 to obtain for $k \geq 0$,

$$\beta_t = \alpha^k - \alpha^{k+1} + (2\gamma n^*)^{-1} \alpha \alpha^k Q(k, a, \alpha, \gamma) + o(d), \quad (3.1)$$

where, $Q(k, a, \alpha, \gamma) = a(1-\alpha)k^2 + (\alpha\gamma - 2\alpha - \gamma - 2a\alpha + 2)k - \alpha(2-\gamma+a)$.

In a similar manner we obtain the β -risk for the case $-1 < k \leq 0$ as

$$\beta_t = 1 - \alpha^{k+1} - (2\gamma n^*)^{-1} a \alpha^{k+1} (k+1)(ak + a - \gamma + 2) + o(d). \quad (3.2)$$

We now turn to compute the fixed sample size β -risk, β_f . Assume that σ is known and we want to establish a $100(1-\alpha)\%$ fixed width confidence interval for μ_0 . Eventually we would use the sample size given in (1.2). Therefore, we would observe as many samples as n^* and compute the interval $I_{n^*} = (X_{n^*(1)} - d, X_{n^*(1)})$ for μ_0 .

Analogously, we write

$$\beta_f = 1 - \alpha^{k+1}, \quad -1 < k \leq 0 \quad (3.3)$$

$$\beta_f = \alpha^k - \alpha^{k+1}, \quad k \geq 0 \quad (3.4)$$

Equations (3.1), (3.2), (3.3) and (3.4), provide the following relation between β -risks of the triple sampling procedure and the corresponding fixed sample size case

$$\beta_t = \beta_f - (2\gamma n^*)^{-1} a \alpha^{k+1} (k+1)(ak + a - \gamma + 2) + o(d), \quad -1 < k \leq 0 \quad (3.5)$$

$$\beta_t = \beta_f + (2\gamma n^*)^{-1} a \alpha^k Q(k, a, \alpha, \gamma) + o(d), \quad k \geq 0 \quad (3.6)$$

Computations based on equations (3.1), (3.2), (3.3) and (3.4) were performed to provide an intuitive appealing feature for the risk associated with both triple sampling and fixed sample size procedures. We took n^* to be 5, 10, 15, 20, 25, 30, 40, 50, 70, 100, 200, 300 and 500, while the parameter γ was set to 0.3, 0.5 and 0.8. The confidence coefficient was set at the usual values 0.90, 0.95 and 0.99. The departure varied over the range $-1 < k \leq 4$. For the

triple sampling risk equations (3.1) and (3.2), each situation was characterized by specifying values for (n^*, γ, α, k) . While for the fixed sample size risk equations (3.3) and (3.4), we specified values for (n^*, α, k) . However, for illustrative purposes we only report the case of $\alpha = 0.05$, $\gamma = 0.5$ and some selected values of k and n^* in Table (I).

To avoid any possible confusion, we emphasize the point that the departure will be to the right of μ_0 (right-departure) when k is negative and to the left of μ_0 (left-departure) when k is positive. This is due to the nature of the alternative hypothesis. Table (I) reveals several points regarding the performance of the triple sampling confidence intervals. For fixed α, γ and $-1 < k \leq 0$, we noticed that β_t increases as n^* increases and approaches β_f as n^* gets large (500). It is also obvious that β_t increases rapidly when n^* is small (≤ 30) and slows down for larger values of n^* . For specific choice of n^* and $-1 < k \leq 0$, we noticed that β_t increases as k increases and attains its maximum value $\approx (1 - \alpha)$ when $k = 0$. This numerical feature supports our findings in equations (3.2) and (3.5). On the contrary, for $0 < k \leq 4$, β_t decreases as k increases and approaches 0 as k approaches 4. The convergence of β_t to 0 occurs faster when n^* gets large. To compare β_t with β_f , we consider the following two subregions. The first covers the interval $0 \leq k \leq k_1$ and the second covers the interval $k_1 \leq k \leq 4$, where k_1 is the solution of $Q(k, a, \alpha, \gamma) = 0$. Values of k_1 are given in Table (II) for some selected values of α and γ . For $0 \leq k \leq k_1$, Table (I) shows that $\beta_t \leq \beta_f$ for all values of n^* and $\beta_t \rightarrow \beta_f$ as n^* gets large. The feature is very much the opposite over the range $k_1 \leq k \leq 4$. It is clear that

β_t decreases as n^* increases and approaches the corresponding β_f as n^* gets large. This is also evident from equations (3.1) and (3.6).

TABLE (I)
 β -risk $\alpha = 0.05, \gamma = 0.5$

k	β_t							β_f
	$n^*=5$	$n^*=10$	$n^*=15$	$n^*=20$	$n^*=30$	$n^*=100$	$n^*=500$	fixed
-0.99	0.021	0.025	0.027	0.027	0.028	0.029	0.029	0.03
-0.80	0.313	0.382	0.405	0.416	0.428	0.444	0.449	0.451
-0.60	0.503	0.601	0.633	0.65	0.666	0.689	0.696	0.698
-0.40	0.638	0.736	0.769	0.785	0.802	0.824	0.832	0.834
-0.20	0.739	0.824	0.852	0.866	0.881	0.9	0.907	0.909
0.00	0.815	0.883	0.905	0.916	0.928	0.943	0.949	0.95
0.50	0.353	0.283	0.259	0.248	0.236	0.219	0.214	0.212
1.00	0.16	0.104	0.085	0.076	0.066	0.053	0.049	0.047
2.00	0.022	0.012	0.009	0.007	0.006	0.003	0.003	0.002
3.00	0.002	0.001	0.001	0.001	0	0	0	0
3.50	0.001	0	0	0	0	0	0	0

TABLE (II)
 Values of (k_1, β)

α	γ		
	0.3	0.5	0.8
0.10	(0.252, 0.504)	(0.265, 0.489)	(0.288, 0.464)
0.05	(0.138, 0.629)	(0.146, 0.614)	(0.161, 0.586)
0.01	(0.036, 0.826)	(0.039, 0.826)	(0.045, 0.806)

Table (II) presents values of k_1 and the associated values of β for different values of α and γ . It is clear from Table (II) that the chance of not detecting

a shift equals to $0.252d$ to the left of μ_0 , for example, is about 50.4% when $\alpha = 0.1$ and $\gamma = 0.3$. Other entries in the table can be interpreted in exactly the same way.

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